

Optimizing MSM registration parameters fixes local minima issues on the dHCP database

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Introduction

MSM (Multimodal Surface Matching) is the state-of-the-art method for spherical registration of cortical surfaces obtained from MRI data. ^{^1} The alignment can be driven by various univariate, multivariate, or multimodal feature sets. This method is commonly used in the community for routine registrations and in particular for data released by the Human Connectome Project (<https://www.humanconnectome.org/>).

The method aligns a source sphere to a target sphere to match their features maps. Registration parameters can be modified according to the requirements of each specific study. Default parameters for sulcal depth is provided. ^{^2} In this configuration, the registration is initialized using a rigid alignment step (rotation of the sphere) to correct global orientation differences between the source and target spheres. It then proceeds through a series of discrete registration steps. During these steps, deformations applied to the source sphere become progressively more local, moving from coarse to fine adjustments.

In this study, using neonatal dHCP data described below, we demonstrate that 1) the default MSM default parameters can provide suboptimal registrations; 2) we quantify the bias induced by such suboptimal registration; 3) we propose an adapted setting of the parameters to address this issue and confirm quantitatively the effectiveness of this approach.

Methods

Data

Right-hemisphere cortical surfaces and corresponding sulcal depth maps were obtained from the Developing Human Connectome Project (dHCP, <https://www.developingconnectome.org/>). The dataset includes 873 neonates aged between 28 and 46 postmenstrual weeks (PMA). For each individual surface, the dHCP also released a parcellation into 17 regions. ^{^3}

The study was approved by the United Kingdom Health Research Authority (Research Ethics Committee reference number: 14/LO/1169) and written parental consent was obtained in every case for imaging and open data release of the anonymized data.

Registrations

Each individual surface was registered to the age-matched template introduced in ^{^4} (<https://brain-development.org/brain-atlases/atlas-from-the-dhcp-project/cortical-surface-templ>

e/) based on sulcal depth, using default MSM parameters setting (Fig. 1.B), as described in the documentation (<https://fsl.fmrib.ox.ac.uk/fsl/docs/#>). The quality of registrations is measured as described below.

Measuring registration accuracy

Sulcal depth maps from the source spheres were resampled onto the template sphere using adaptive barycentric interpolation (MSM utility). Pearson correlation coefficients were computed for each subject between the resampled maps and the corresponding template features. Jacobians were computed as the integral of the base-2 logarithm of the relative ratio between the face areas of the registered and source spheres, propagated to each vertex.

We merged selectively some small parcels from the provided parcellations in order to obtain 7 regions of similar area covering the entire hemisphere surface. The resulting parcellation is illustrated on Fig. 1.A. Parcellation maps were resampled onto the template sphere using nearest-neighbor interpolation. For each parcel, we then computed the Dice coefficient for all possible pairs of registered subjects. For each subject, a single score was obtained by averaging the values across regions and pairs of subjects. We also computed the Pearson correlation between the registered and target spheres, and the Jacobians between the source and registered spheres in each parcel.

We finally computed density maps for the region values by summing the values at each vertex and normalizing between 0 and 1.

New MSM parameter settings

We assume that MSM can converge towards local minima, as any other registration technique based on gradient descent. We expect suboptimal registration to occur more often for older subjects with complex folding patterns. We investigate alternative settings of MSM allowing us to achieve better registration for all the individuals, by improving the robustness of the rigid transformation (rotation) serving as an initialization for subsequent discrete, local deformations.

We selected 50 subjects with suboptimal registrations based on Jacobian values much larger than the average of the population. For these subjects, we performed a series of rigid registrations. We varied the value of the sigma parameter, which controls kernel smoothing for the source and template sulcal depth maps from 2 to 100 by steps of 5 to identify the value that optimizes registration across all subjects.

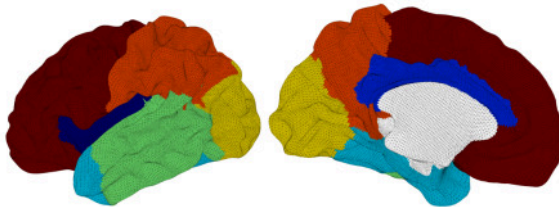
Standard MSM with additional rigid initialization was applied to all outliers using optimal sigma (Fig. 2.B).

We performed paired Wilcoxon signed-rank and Mann-Whitney U tests to statistically compare the mean Jacobian determinant, Dice coefficient, and Pearson correlation values obtained using either the default or our optimized MSM settings between subjects with high and low registration performance. Effect sizes were computed, and statistical significance was set at $\alpha = 0.05$. Cluster 1 was randomly undersampled ($n = 261$) between 35 and 44 PMA for comparison. We performed an FDR correction for vertex-wise tests across subjects.

Methods Figure (Optional)

Fig.1.

A.



B.

```
--lambda=0,0,0.1,0.2,0.3
--opt=AFFINE,AFFINE,DISCRETE,DISCRETE,DISCRETE
--it=50,50,3,3,3
--sigma_in=σ,2,2,3,2
--sigma_ref=σ,2,2,1.5,1
--datagrid=4,4,4,5,6
--CPgrid=0,0,2,3,4
--SGgrid=0,0,4,5,6
--IN
```

A. Merged parcellation map projected onto the 43 PMA atlas. The parcellation of a 43 PMA subject was resampled and mapped to the corresponding age template via nearest neighbors interpolation. Twelve labels were aggregated into external temporal, internal temporal, and limbic regions. B. MSM configuration file. Green: standard MSM parameters; yellow: proposed additional rigid initialization parameters; green+yellow: MSM with additional rigid initialization. σ : the parameter value adjusted in rigid registration sessions. Comma-separated lists define parameters per level. The number of resolution levels in msm equals the list length (for more details, see the FSL documentation).

Results

Registrations with default setting of the parameters

The distributions of Dice, correlation, and Jacobian measures are bimodal, indicating the presence of two classes (referred as Cluster 1 and Cluster 2 below) of registration performance. 30% ($n = 261$; Cluster 1) of the registrations have low correlation values ($\text{corr} < 0.7$; $\text{Dice} < 0.8$) and high Jacobians (> 8), whereas 70% ($n = 612$; Cluster 2) have high correlation values ($\text{corr} > 0.7$; $\text{Dice} > 0.8$) and low Jacobians (< 8) (Fig. 2.A).

This trend is also observed at the parcels level for Dice values and Jacobians, although some regions are better preserved and easier to align than others.

The subjects in cluster 2 are aged between 38 and 44 PMA, and the subjects in cluster 1 are distributed across the entire age range.

The density map reveals a mismatch in the registered region labels across all subjects when using default MSM parameters largely driven by Cluster 2 registrations, similar to the unregistered source regions (Fig. 2.E).

A Wilcoxon-Mann-Whitney test between Cluster 1 and Cluster 2 for the MSM default settings shows significant differences in the distributions of Jacobians, Dice scores, and Pearson correlation coefficients ($p < 0.05$; effect size > 0.3) (Fig. 2.D).

Optimization of sigma for robust rigid registration

The global optimum is reached when sigma for both the source and template is 20. This value corresponds to the point at which the correlation is maximized while minimizing distortions across all 50 subjects (Fig. 2.B).

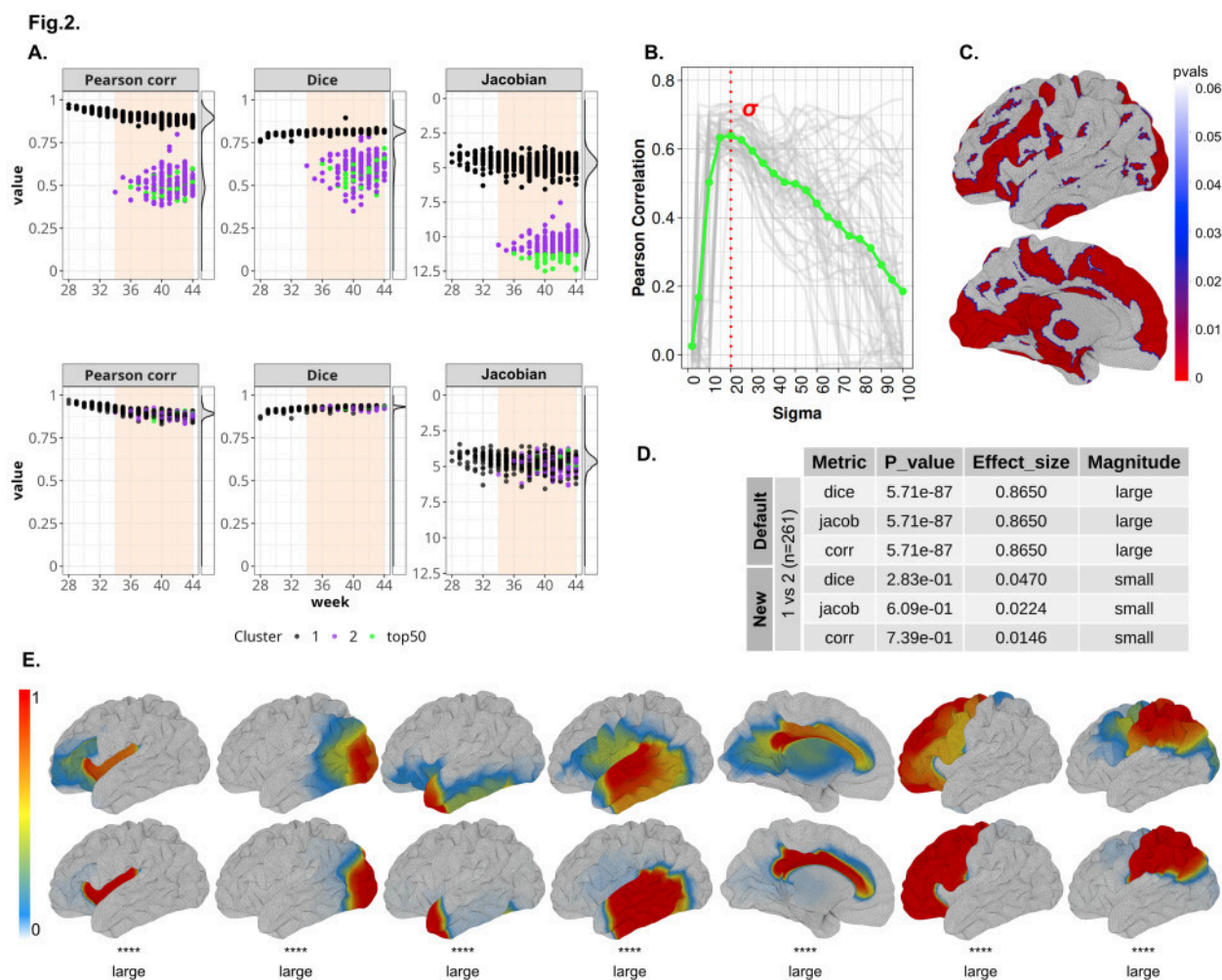
Registrations with new MSM default parameters

For all cortical surfaces in Cluster 2, a paired Wilcoxon test on the Jacobian values at each vertex shows a significant difference between the default settings and the optimized parameters ($p < 0.05$; effect size > 0.3 ; FDR correction) (Fig. 2.C).

The Mann-Whitney U test comparing Cluster 1 and Cluster 2 under the optimized settings shows no significant differences in the distributions of Jacobians, Dice scores, and Pearson correlation coefficients ($p > 0.05$; effect size < 0.3 ; Fig. 2.A and Fig. 2.D).

Visual inspection of the density maps of the parcellations shows equivalent and consistent patterns across conditions. A paired Wilcoxon signed-rank test performed for each parcel on Dice values across all subjects, comparing the default and new MSM settings, shows a significant difference between the distributions ($p < 0.05$; effect size > 0.3 ; Fig. 2.E).

Results Figure (Optional)



Conclusion

This study highlights the limitations of the default settings of the parameters for the widely used MSM registration technique. By simply adding an additional rigid registration step based on smoothed sulcal depth maps for both source and reference surfaces, we demonstrate that registration quality can be significantly improved. Integrated into the full pipeline, this step increases correlation, reduces distortions, and homogenizes Dice distributions across all outliers.

These results show the relative robustness of MSM to inter-subject variability and parameter selection. It highlights the need for systematic quality control, using global and local metrics, to identify registration failures. In the neonatal context, these biases might affect the interpretation of developmental trajectories and regional morphometric analyses.

References/Citations

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Modeling and Analysis Methods: Image Registration and Computational Anatomy

Secondary Parent Category & Sub-Category

Lifespan Development: Peri- and Neo-natal Development

Keywords

MRI, Data Registration, Development, Informatics, Computing, Cortex, Other

If Other, please list the terms below. Multiple terms must be separated by semi-colons (;).

Multimodal Surface Matching (MSM)